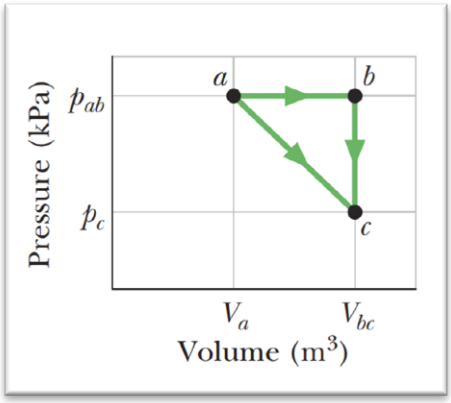
PART II: KINETIC THEORY OF GASES (PART 2)

1 A hydrogen molecule (diameter 1.0 x 10^{-8} cm), traveling at the rms speed, escapes from a 4000K furnace into a chamber containing cold argon atoms (diameter 3.0 x 10^{-8} cm) at a density of 4.0 x 10^{19} atoms/cm^3. (a) What is the root-mean-square speed of the hydrogen molecule? (b) If it collides with an argon atom, what is the closest their centers can be, considering each as spherical? (c) What is the initial number of collisions per second experienced by the hydrogen molecule?

2 What is the internal energy of 1.0 mol of an ideal monatomic gas at 273 K?

3 One mole of an ideal diatomic gas goes from a to c along the diagonal path **in Fig. below**. The scale of the vertical axis is set by p_{ab} = 5.0 kP$_a$ and p_c = 2.0 kP$_a$, and the scale of the horizontal axis is set by V_{bc} = 4.0 m^3 and V_a = 2.0 m^3. During the transition, (a) what is the change in internal energy of the gas, and (b) how much energy is added to the gas as heat? (c) How much heat is required if the gas goes from a to c along the indirect path *abc*?



4 A container holds a mixture of three nonreacting gases: 2.40 mol of gas 1 with C_{V1} 12.0 J/mol·K 1.50 mol of gas 2 with C_{V2} =12.8 J/mol·K, and 3.20 mol of gas 3 with C_{V3} = 20.0 J/mol·K. What is C_V of the mixture?

5 We give 70 J as heat to a diatomic gas, which then expands at constant pressure. The gas molecules rotate but do not oscillate. By how much does the internal energy of the gas increase?

6 Suppose 12.0 g of oxygen (O_2) gas is heated at constant atmospheric pressure from 25.0^{0}C to 125^{0}C. (a) How many moles of oxygen are present? (See Table below for the molar mass.) (b) How much energy is transferred to the oxygen as heat? (The molecules rotate but do not oscillate.) (c) What fraction of the heat is used to raise the internal energy of the oxygen?

Table 19-1 Some RMS Speeds at Rc Temperature ($T = 300$ K)[a]

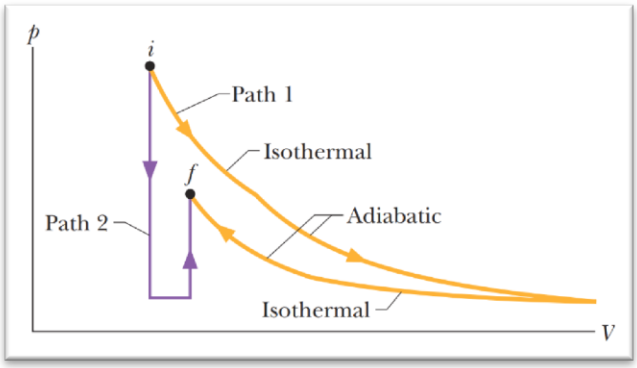
Gas	Molar Mass (10^{-3} kg/mol)	v_{rms} (m/s)
Hydrogen (H_2)	2.02	1920
Helium (He)	4.0	1370
Water vapor (H_2O)	18.0	645
Nitrogen (N_2)	28.0	517
Oxygen (O_2)	32.0	483
Carbon dioxide (CO_2)	44.0	412
Sulfur dioxide (SO_2)	64.1	342

[a]For convenience, we often set room temperature equal to 300 K even though (at 27°C or 81°F) that represents a fairly warm room.

7 Suppose 1.00 L of a gas with γ = 1.30, initially at 273 K and 1.00 atm, is suddenly compressed adiabatically to half its initial volume. Find its final (a) pressure and (b) temperature. (c) If the gas is then cooled to 273 K at constant pressure, what is its final volume?

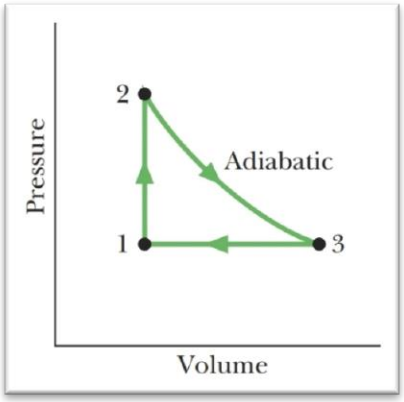
8 *Opening champagne*. In a bottle of champagne, the pocket of gas (primarily carbon dioxide) between the liquid and the cork is at pressure of *pi* = 5.00 atm. When the cork is pulled from the bottle, the gas undergoes an adiabatic expansion until its pressure matches the ambient air pressure of 1.00 atm. Assume that the ratio of the molar specific heats is $\gamma = \frac{4}{3}$. If the gas has initial temperature T_i = 5.00^{0}C, what is its temperature at the end of the adiabatic expansion?

9 Figure below shows two paths that may be taken by a gas from an initial point *i* to a final point *f*. Path 1 consists of an isothermal expansion (work is 50 J in magnitude), an adiabatic expansion (work is 40 J in magnitude), an isothermal compression (work is 30 J in magnitude), and then an adiabatic compression (work is 25 J in magnitude). What is the change in the internal energy of the gas if the gas goes from point *i* to point *f* along path 2?



10 An ideal diatomic gas, with rotation but no oscillation, undergoes an adiabatic compression. Its initial pressure and volume are 1.20 atm and 0.200 m^3. Its final pressure is 2.40 atm. How much work is done by the gas?

11 Figure below shows a cycle undergone by 1.00 mol of an ideal monatomic gas. The temperatures are T_1 = 300 K, T_2 = 600 K, and T_3 = 455 K. For 1 : 2, what are (a) heat Q, (b) the change in internal energy ΔE_{int}, and (c) the work done W? For 2 : 3, what are (d) Q, (e) ΔE_{int}, and (f) W? For 3 : 1, what are (g) Q, (h) ΔE_{int}, and (i) W? For the full cycle, what are (j) Q, (k) ΔE_{int}, and (l) W? The initial pressure at point 1 is 1.00 atm (= 1.013 x 10^5 Pa). What are the (m) volume and (n) pressure at point 2 and the (o) volume and (p) pressure at point 3?



MCQs

1. For an ideal diatomic gas:

a. C_v = 20.8 J/mol.K
b. C_v = 28.5 J/mol.K
c. C_v = 20.8 J/K
d. C_v = 25.5 J/K

2. A diatomic gas, expanded isobarically, did 42 J of work. Which was the amount of heat given to the gas during such process?

a. 147 J
b. 137 J
c. 127 J
d. 117 J

3. 1.80 mol of an ideal gas is taken from a volume of 3.00 m^3 to a volume of 1.50 m^3 via an isothermal compression at 30°C. How much energy is transferred as heat during the compression?

a. – 3140 J
b. – 3.14 J
c. – 314 J
d. – 31.4 J

4. The adiabatic relation between volume and temperature of an ideal gas is:

a. $V^{\gamma-1}\,T$ = constant
b. $V^{1+\gamma}\,T^{\gamma}$ = constant
c. $V^{1-\gamma}\,T^{\gamma+1}$ = constant
d. $V^{1-\gamma}\,T^{\gamma-1}$ = constant

5. Considering an ideal gas, which of the following statements is correct?

a. C_p can never be less than C_v
b. C_p is always twice C_v
c. C_p may be less than C_v depending on the gas
d. C_p may be equal to or less than C_v

6. For a diatomic gas, the value of $\gamma = \frac{C_p}{C_v}$ is:

a. $\frac{9}{7}$
b. $\frac{7}{5}$
c. $\frac{5}{3}$
d. $\frac{5}{7}$

7. The work done by an adiabatic expansion is written as:

a. $W = \frac{P_2V_2 - P_1V_1}{1-\gamma}$

b. $W = 0$

c. $W = \frac{P_2V_2 - P_1V_1}{\gamma - 1}$

d. $W = P_2V_2 \ln(\frac{P_1}{P_2})$

8. A monatomic gas is suddenly compressed to $(\frac{1}{8})^{th}$ of its initial volume adiabatically. The ratio of its final pressure to the initial pressure is

a. 32

b. 8

c. $\frac{1}{32}$

d. $\frac{1}{8}$

9. One mole of a monoatomic ideal gas is mixed with one mole of a diatomic ideal gas. The molar specific heat of the mixture at constant volume is:

a. R

b. $2R$

c. 0.5 R

d. 4R

10. A gaseous mixture consists of 16g of Helium and 16g of Oxygen. The ratio $\frac{C_p}{C_v}$ of the mixture is:

a. 1.4

b. 1.54

c. 1.59

d. 1.62